

August 11, 2026

Longitudinal Dynamics and Grid Tomography

Cris

[#ai-haai-beam-tomography](#)



U.S. DEPARTMENT
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01

Longitudinal Dynamics



Two Different Approaches for Accelerator with RF

Hamiltonian Equations

Smooth Distributed RF around the Ring

$$\mathcal{H}(\phi, \Delta E) = \frac{\eta_0}{2\beta_s^2 E_s} (\Delta E)^2 + \frac{qV}{2\pi h} [\cos \phi - \cos \phi_s + (\phi - \phi_s) \sin \phi_s]$$

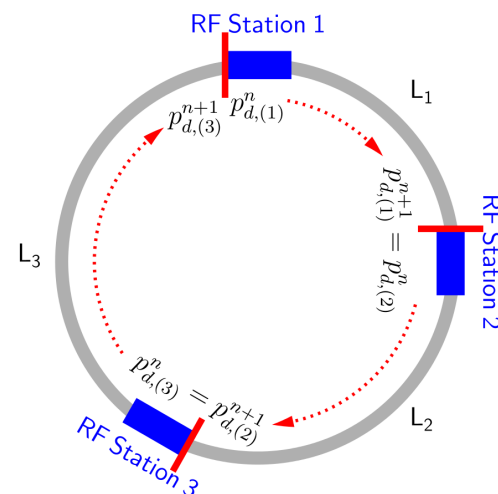
RF Stationary
Buckets

$$\phi_s = \pi, \quad \sin \phi_s = 0, \quad \cos \phi_s = -1$$

$$\mathcal{H} = \frac{P^2}{2} + 1 - \cos \theta$$

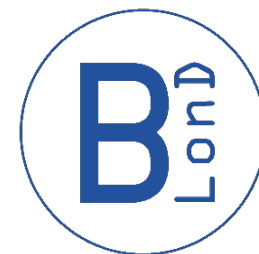
Pendulum Equation

Symplectic Mappings



$$\Delta E_{n+1} = \Delta E_n + qV_n [\sin (\omega_{\text{rf},n} \Delta t_n + \phi_{\text{RF},n}) - \sin \phi_{s,n}]$$

$$\Delta t_{n+1} = \Delta t_n + \frac{\eta_{0,n+1} T_{0,n+1}}{\beta_{s,n+1}^2 E_{s,n+1}} \Delta E_{n+1}$$

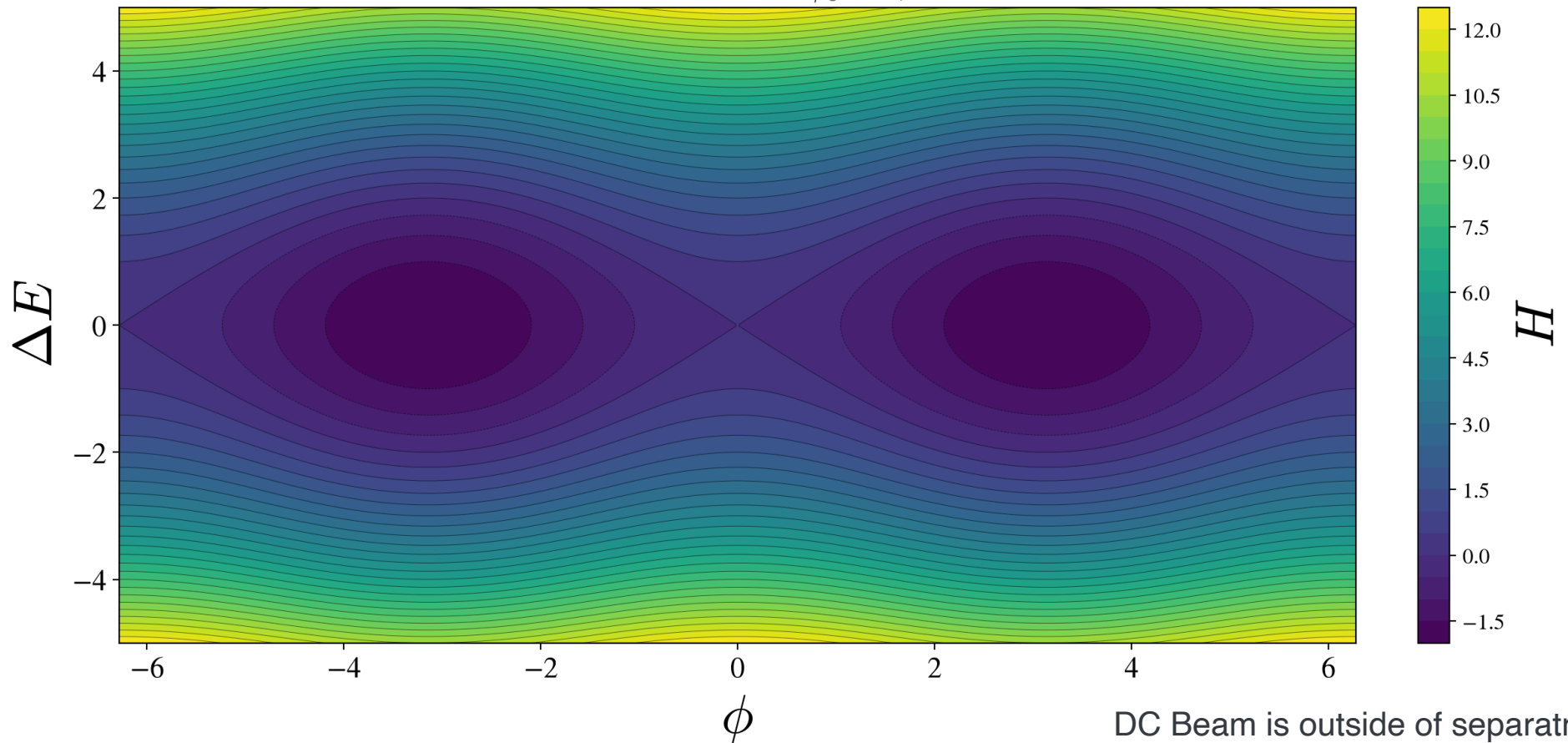




Hamiltonian Mechanics

Two distinct regions: Inside the separatrix (Bounded Motion Libration), outside (Rotation)

RF Hamiltonian contours: $\psi_s = \pi$, $\Omega = 1.0$



DC Beam is outside of separatrix



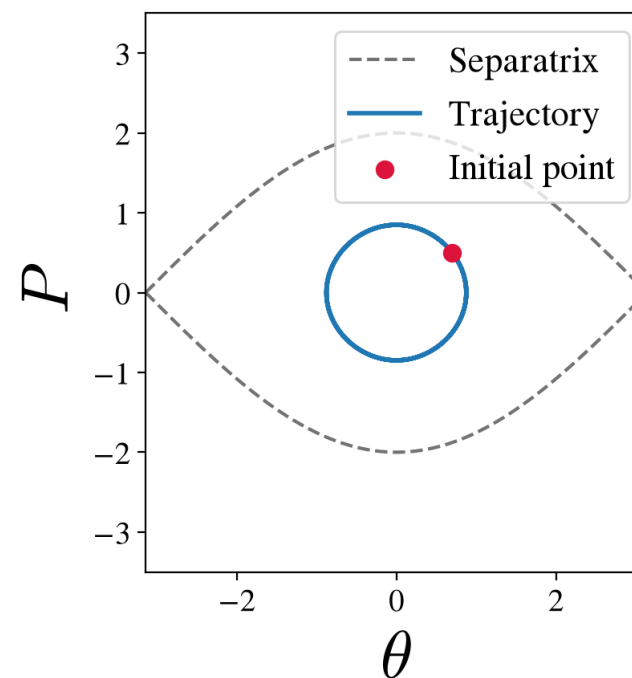
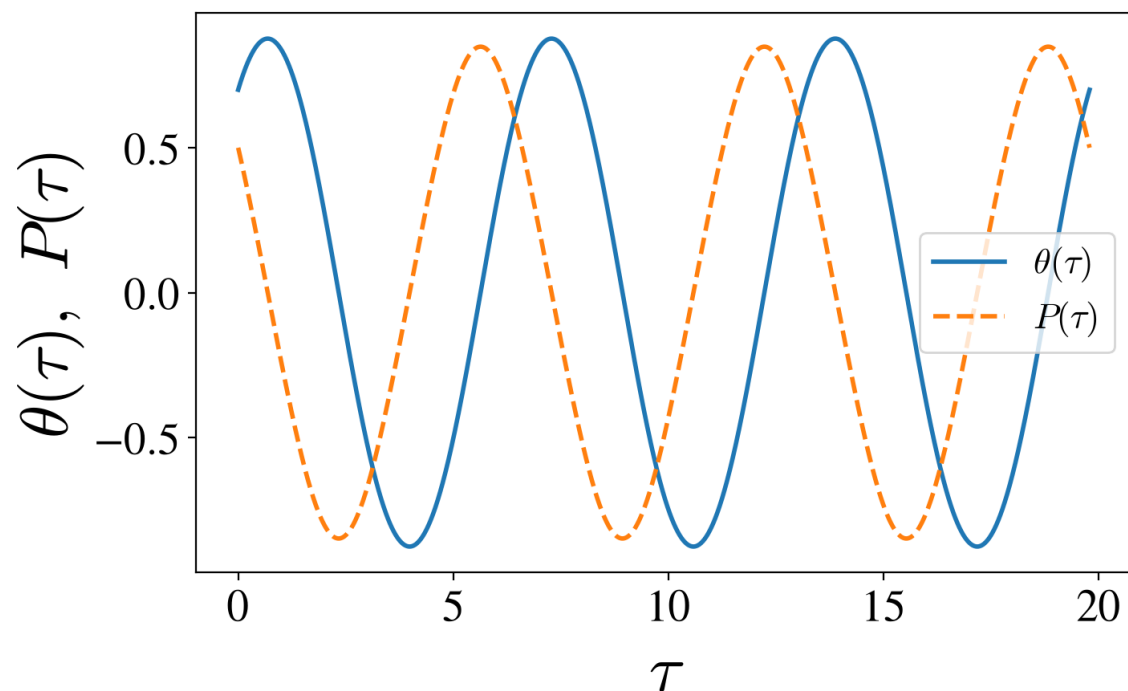
Libration (Bounded Motion)

$$k_b^2 = \frac{1 + \varepsilon_b}{2}, \quad 0 < k_b < 1$$

$$\theta(\tau) = 2 \arcsin[k_b \operatorname{sn}(z_b(\tau), k_b)]$$

$$P(\tau) = 2\sigma k_b \operatorname{cn}(z_b(\tau), k_b)$$

Libration | E=-0.63984 | modulus=0.42436 | period=6.59887



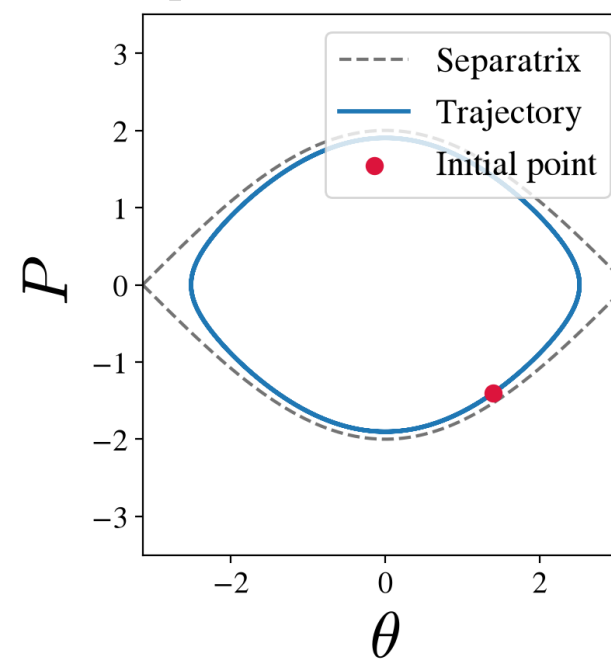
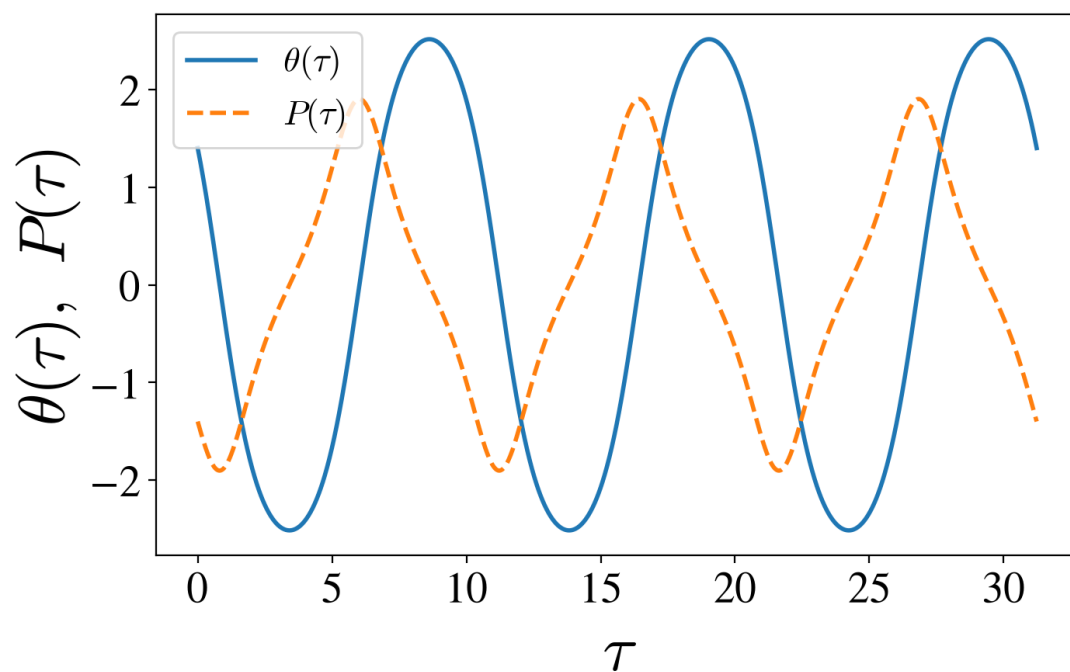


Libration (Bounded Motion)

$$\theta(\tau) = 2 \arcsin[k_b \operatorname{sn}(z_b(\tau), k_b)]$$

$$P(\tau) = 2\sigma k_b \operatorname{cn}(z_b(\tau), k_b)$$

Libration | E=0.81003 | modulus=0.95132 | period=10.40934



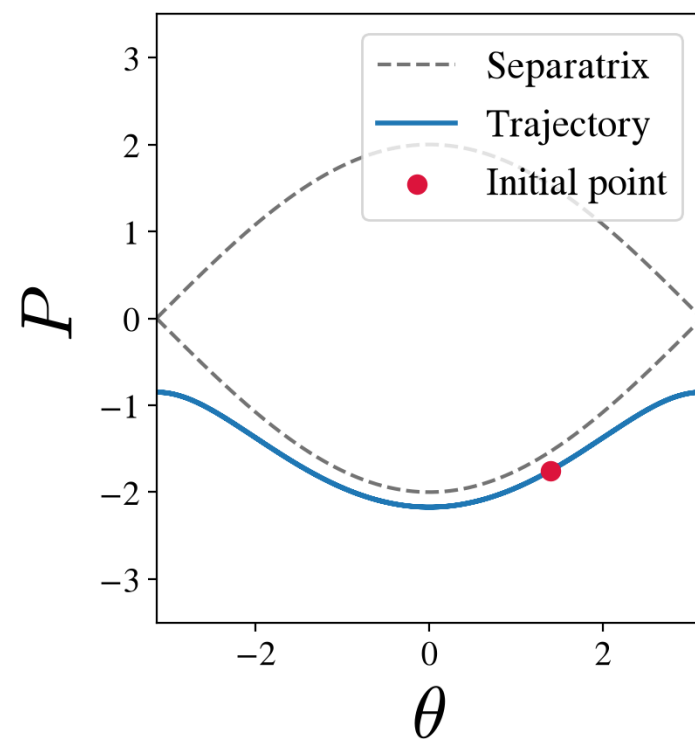
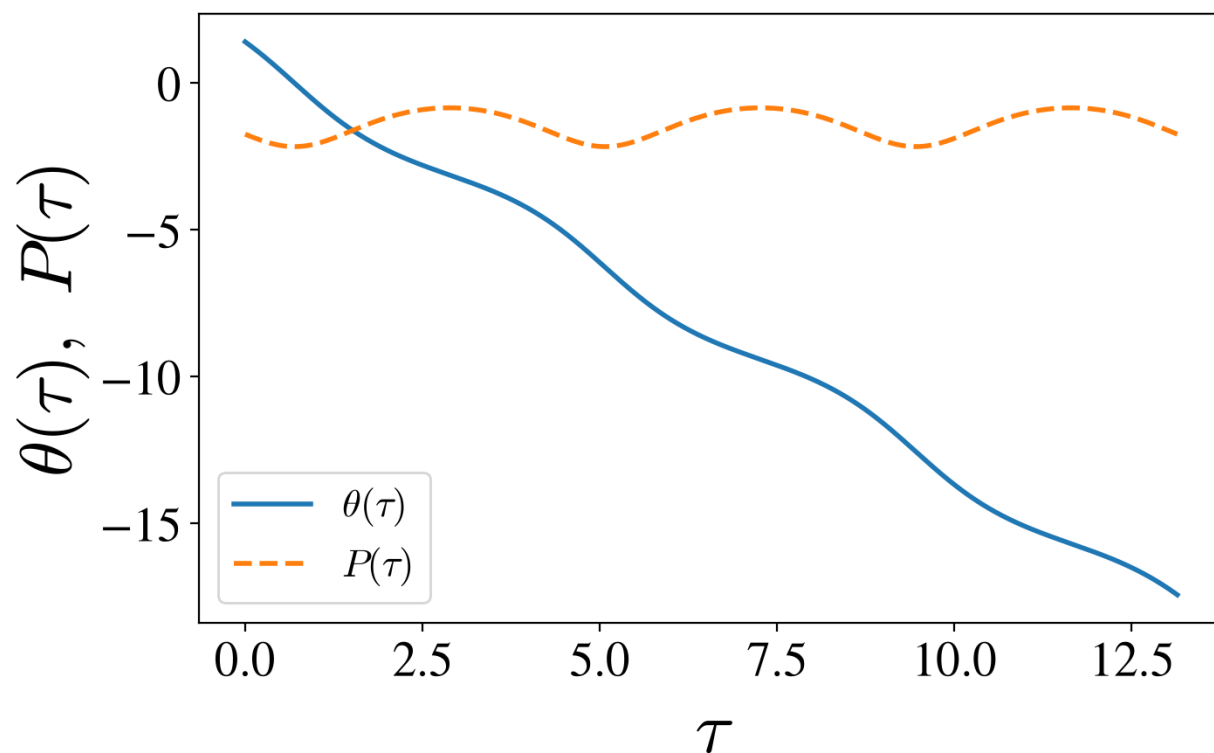


Rotation (Unbounded Motion)

$$\theta(\tau) = 2 \operatorname{am}(z_u(\tau), k_u)$$

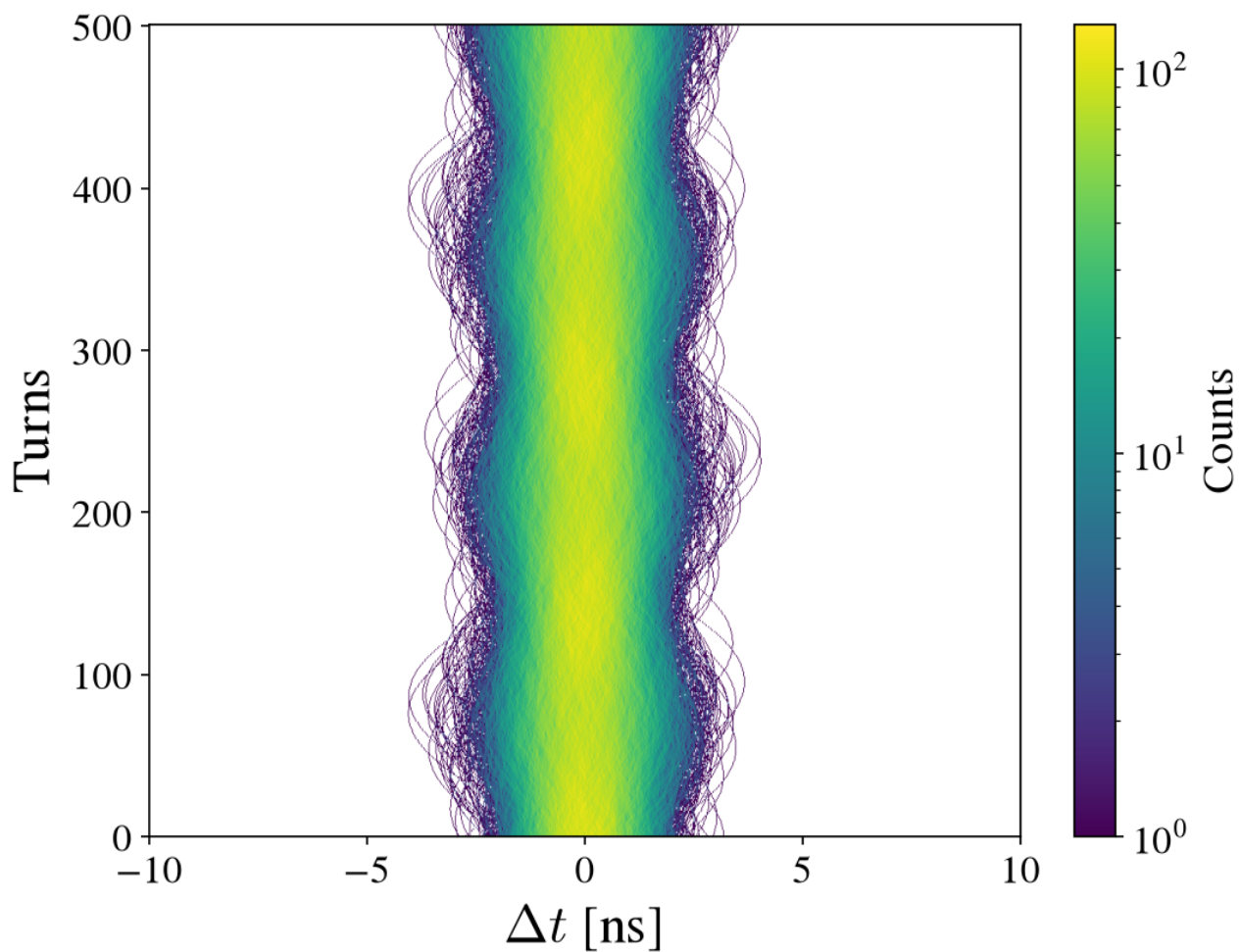
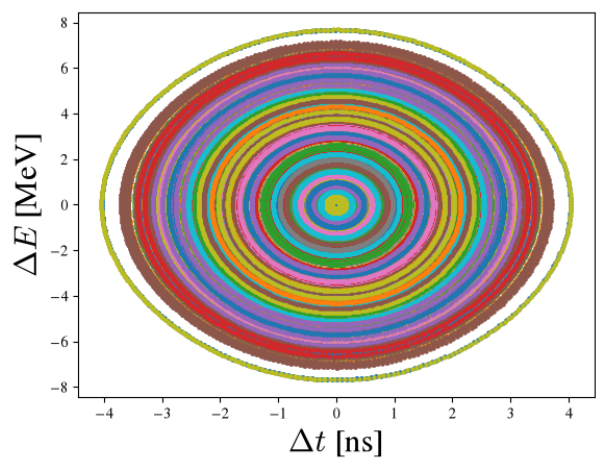
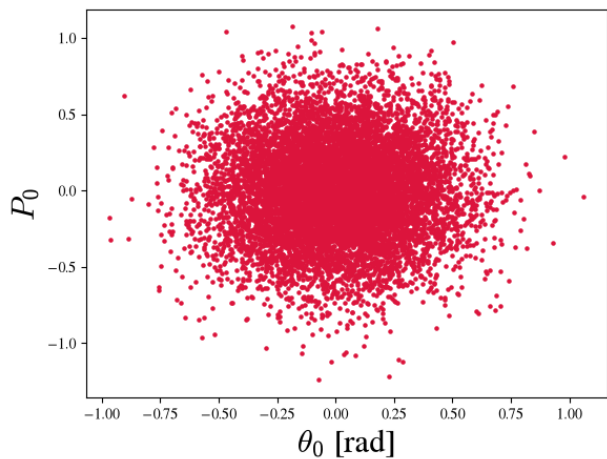
$$P(\tau) = \sigma \sqrt{2(\varepsilon_u + 1)} \operatorname{dn}(z_u(\tau), k_u)$$

Rotation | E=1.36128 | modulus=0.92032 | period=4.38050





Resistive Wall Current Monitor (Projection on time or phase)





Reference on Slip Stacking Dynamics (Recycler Ring)

PHYSICAL REVIEW ACCELERATORS AND BEAMS **20**, 064202 (2017)

Hamiltonian approach to slip-stacking dynamics

S. Y. Lee

Department of Physics, Indiana University, Bloomington, Indiana 47405, USA

K. Y. Ng

Fermi National Accelerator Laboratory, Batavia, Illinois 60510, USA

(Received 25 March 2017; published 29 June 2017)

Hamiltonian dynamics has been applied to study the slip-stacking dynamics. The canonical-perturbation method is employed to obtain the second-harmonic correction term in the slip-stacking Hamiltonian. The Hamiltonian approach provides a clear optimal method for choosing the slip-stacking parameter and improving stacking efficiency. The dynamics are applied specifically to the Fermilab Booster-Recycler complex. The dynamics can also be applied to other accelerator complexes.

DOI: [10.1103/PhysRevAccelBeams.20.064202](https://doi.org/10.1103/PhysRevAccelBeams.20.064202)

PHYSICAL REVIEW ACCELERATORS AND BEAMS **19**, 104001 (2016)

Enhanced dynamical stability with harmonic slip stacking

Jeffrey Eldred and Robert Zwaska

Fermi National Accelerator Laboratory, Batavia, Illinois 60510, USA

(Received 22 July 2016; published 26 October 2016)

We develop a configuration of radio-frequency (rf) cavities to dramatically improve the performance of slip stacking. Slip stacking is an accumulation technique used at Fermilab to nearly double proton intensity by maintaining two beams of different momenta in the same storage ring. The two particle beams are longitudinally focused in the Recycler by two 53 MHz 100 kV rf cavities with a small frequency difference between them. We propose an additional 106 MHz 20 kV rf cavity with a frequency at the double the average of the upper and lower main rf frequencies. We show the harmonic rf cavity cancels out the resonances generated between the two main rf cavities and we derive the relationship between the harmonic rf voltage and the main rf voltage. We find the area factors that can be used to calculate the available phase space area for any set of beam parameters without individual simulation. We establish Booster beam quality requirements to achieve 99% slip stacking efficiency. We measure the longitudinal distribution of the Booster beam and use it to generate a realistic beam model for slip stacking simulation. We demonstrate that the harmonic rf cavity can not only reduce particle loss during slip stacking, but also reduce the final longitudinal emittance.

DOI: [10.1103/PhysRevAccelBeams.19.104001](https://doi.org/10.1103/PhysRevAccelBeams.19.104001)



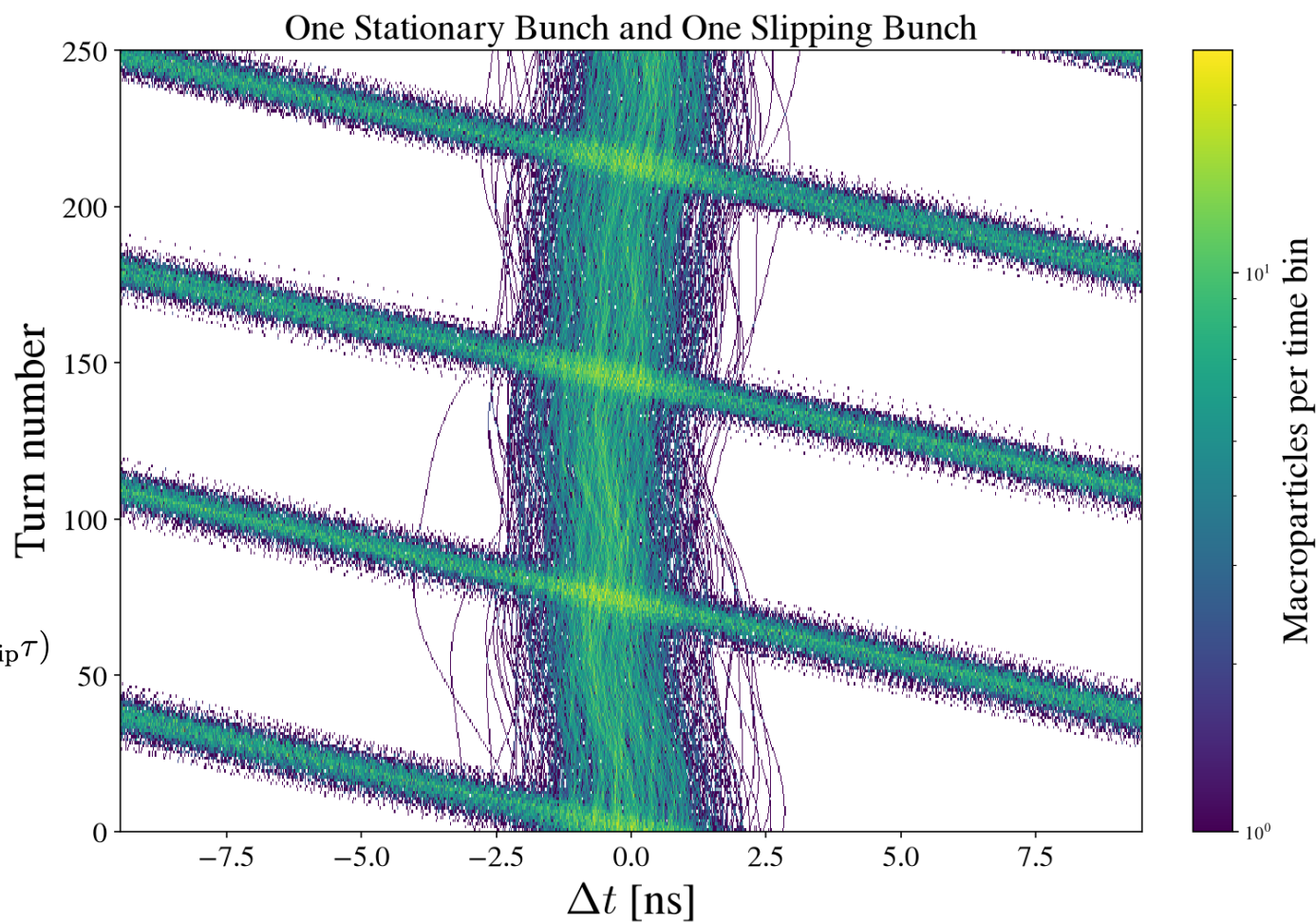
Resistive Wall Current Monitor (Projection on time or phase)

Stationary Bucket

$$H(\theta, P) = \frac{P^2}{2} - \cos \theta$$

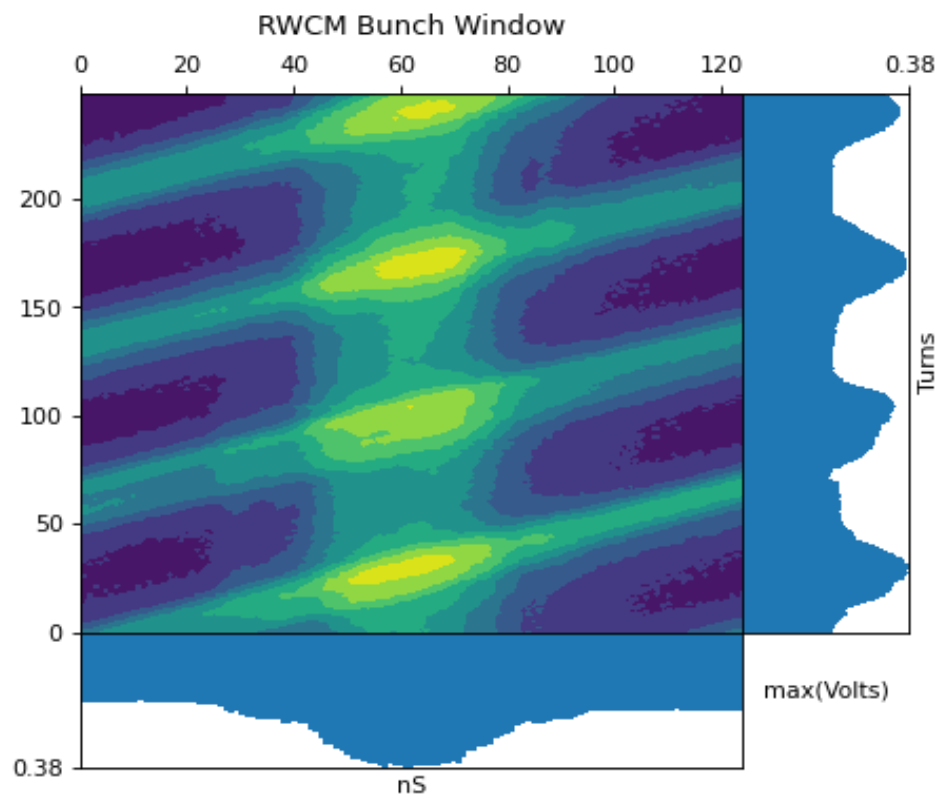
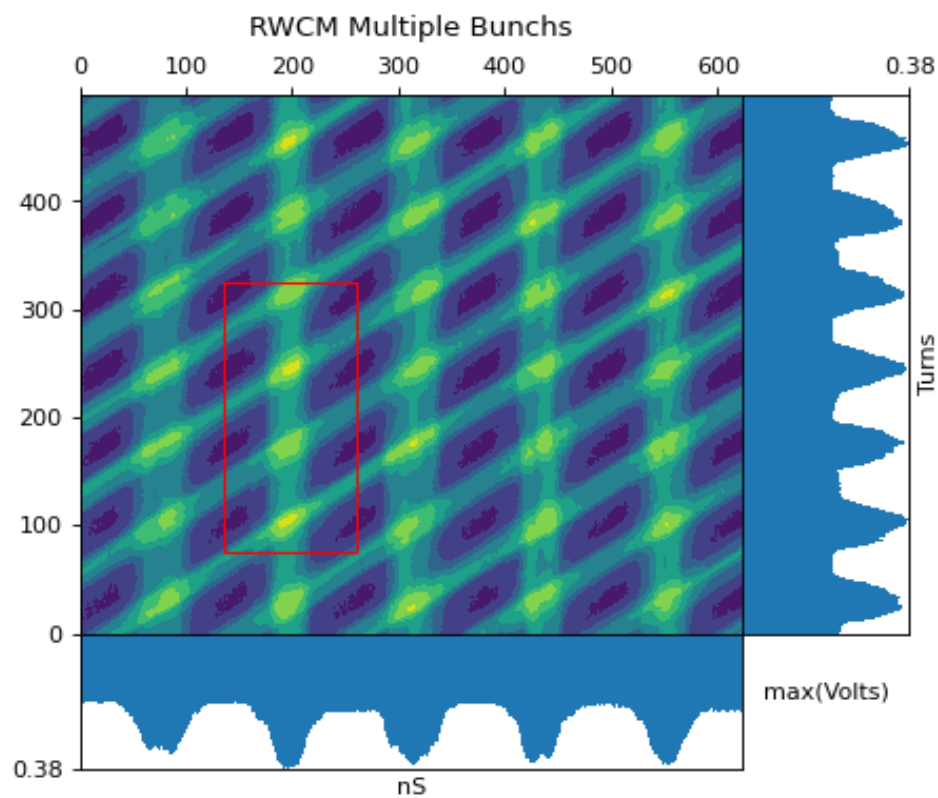
Slipping Bunch

$$H(\theta, P) = \frac{(P - \Delta P_{\text{sep}})^2}{2} - \cos(\theta - \nu_{\text{slip}}\tau)$$





Resistive Wall Current Monitor Experimental (Kyle and Kit)





02

Grid Tomography



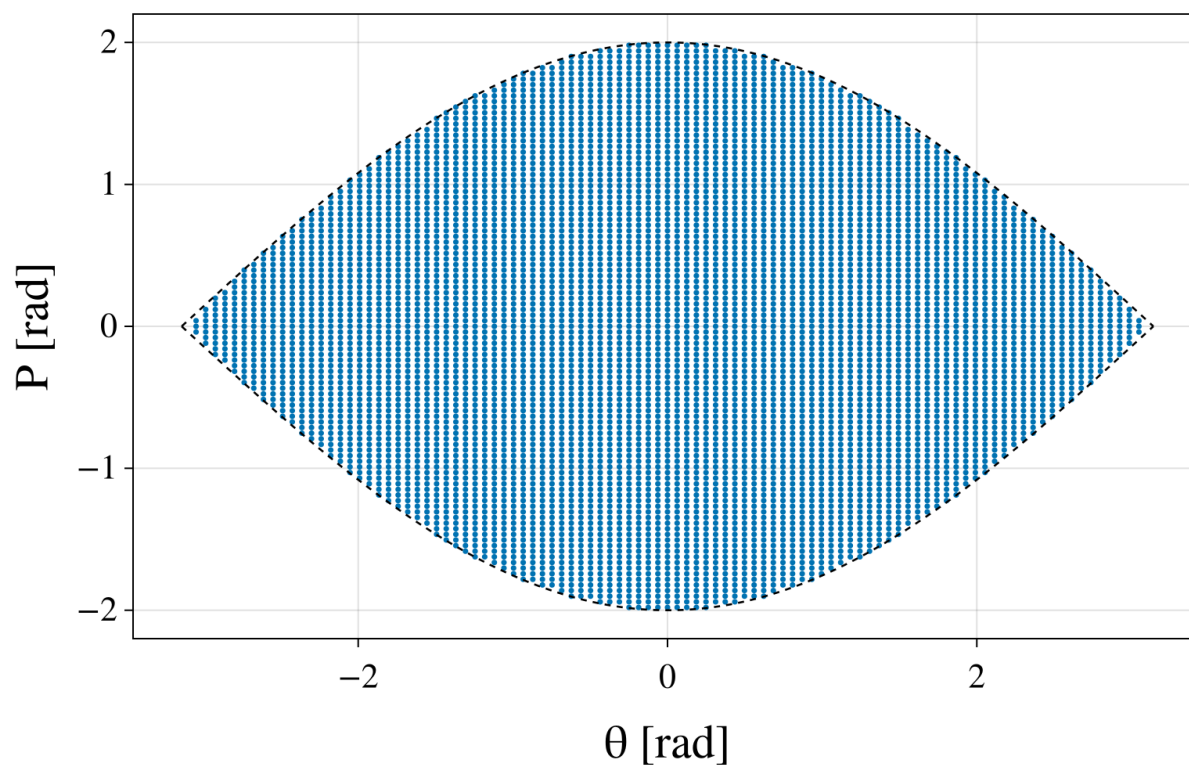
Grid Tomography

Each grid point will have a cell mass

$$\hat{R}_{nb}(\mathbf{a}) = \sum_{j=1}^{N_c} A_{nb,j} w_j(\mathbf{a})$$

$$\hat{\mathbf{R}} = A\mathbf{w}$$

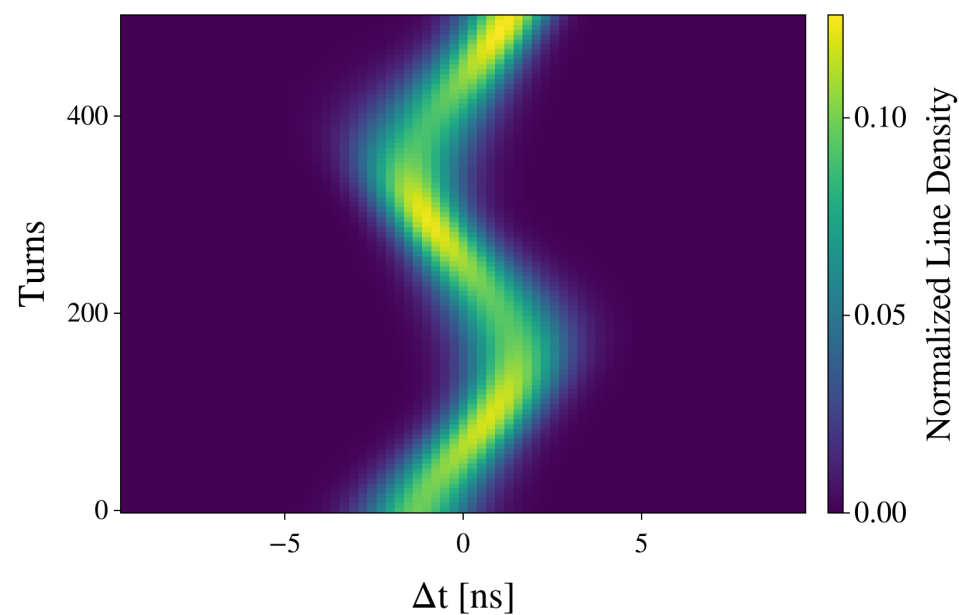
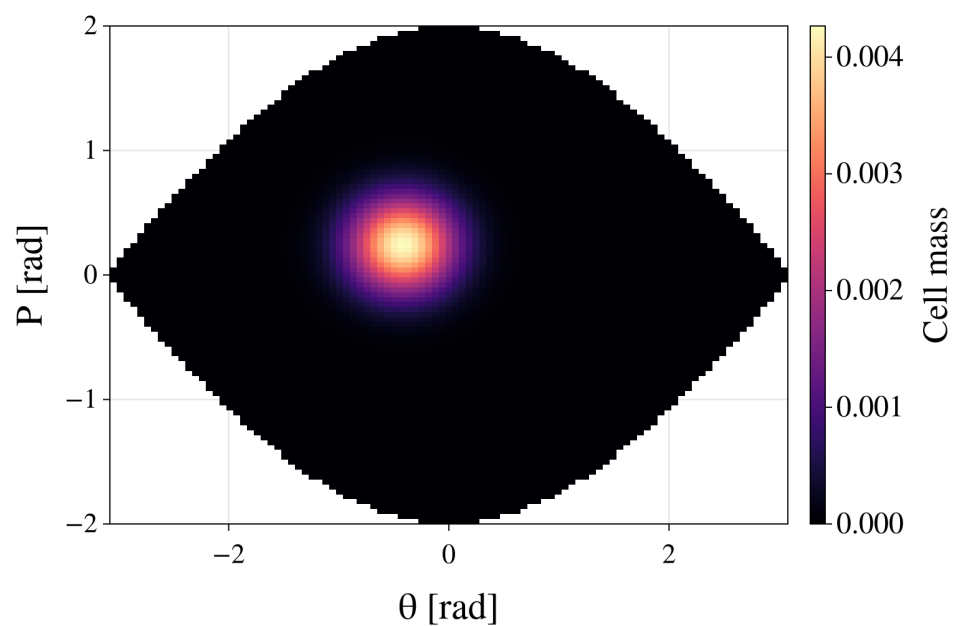
$$\rho(\theta, P) = \sum_j^{N_G} w_j \delta(\theta - \theta_j) \delta(P - P_j)$$





Synthetic Ground Truth (Target)

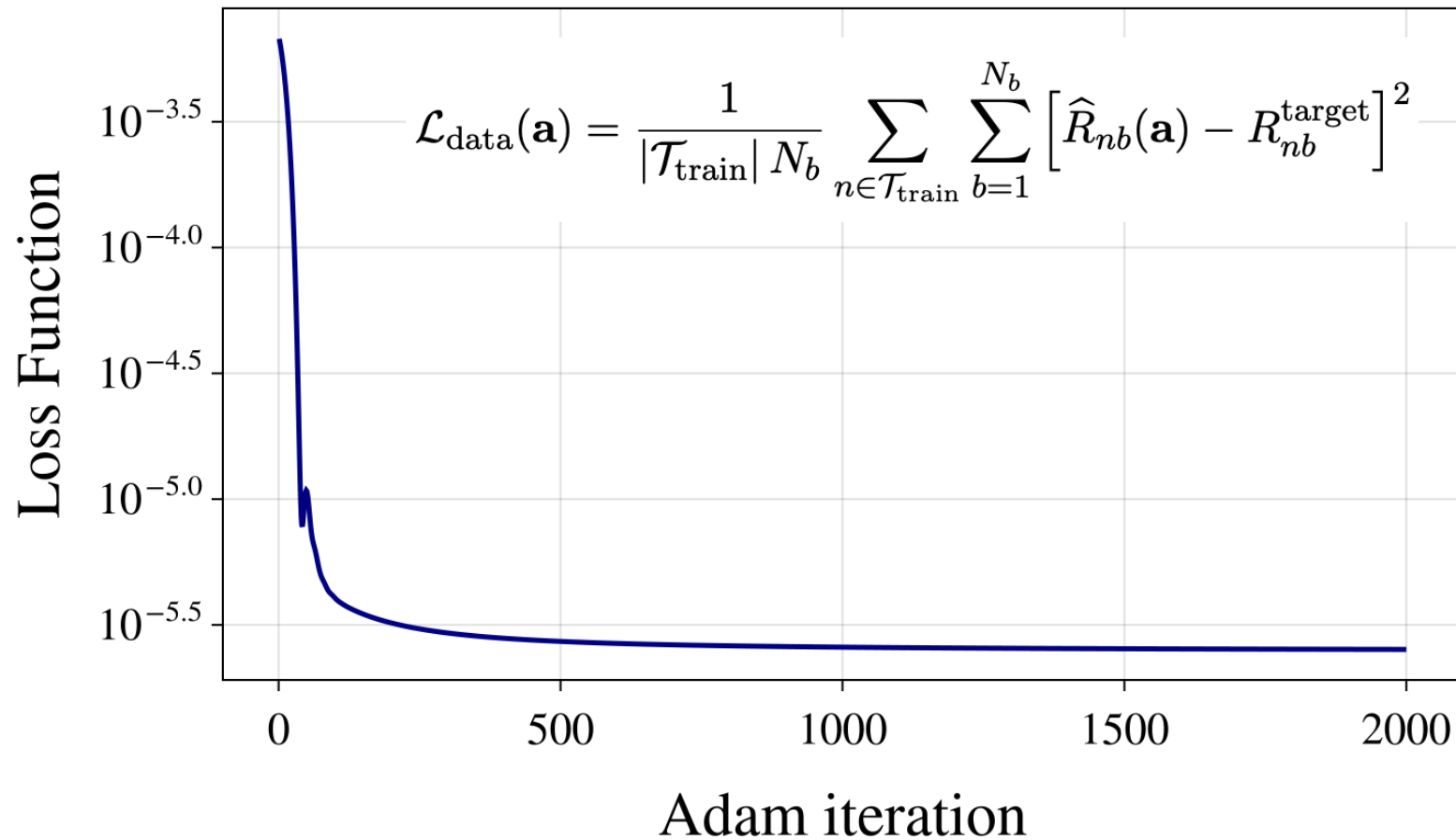
Cell masses are distributed with a 2D Gaussian at every point in grid. Ground truth RWCM.





Loss Function Minimization

MSE is minimized using ADAM optimizer through Julia differentiable package

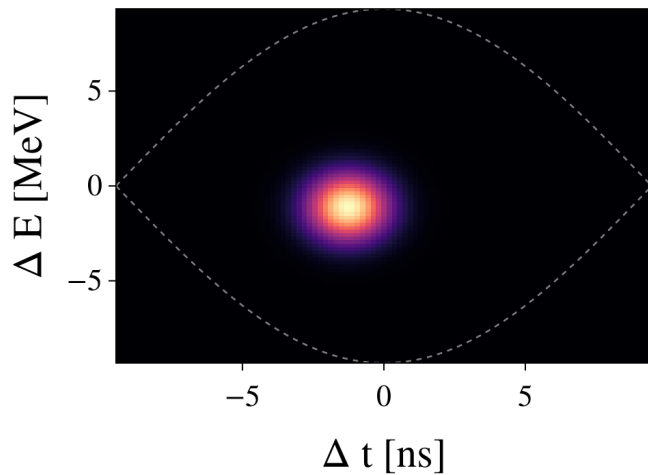




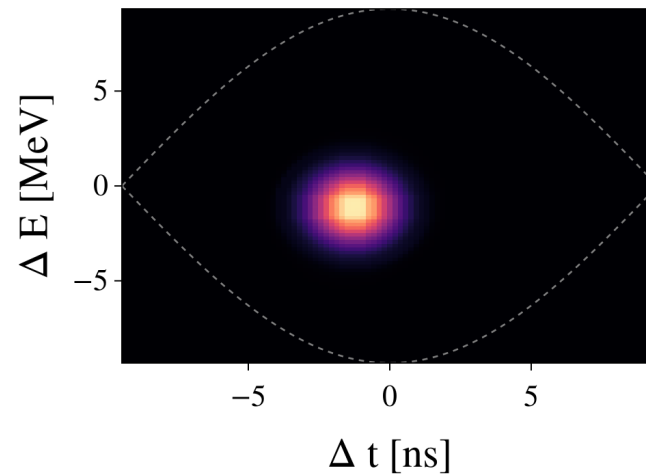
Gaussian Off-Center Distribution

Each grid point will have a cell mass

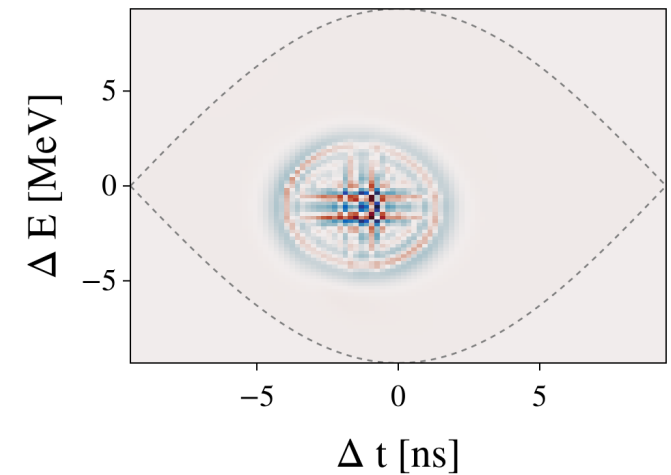
Ground-truth initial density



Grid-tomography reconstruction



Residuals

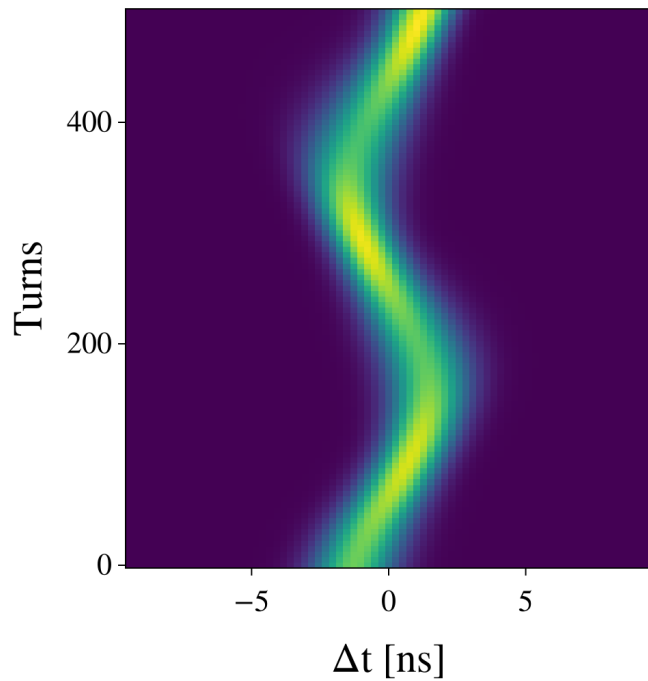




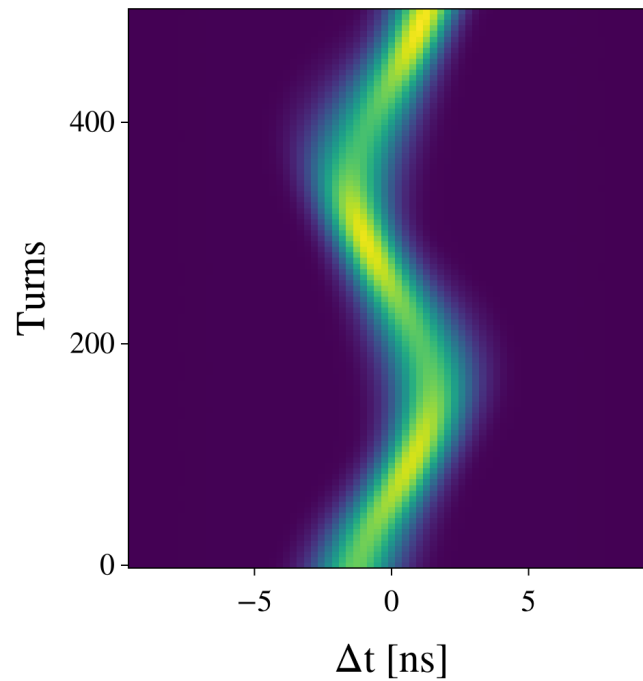
Gaussian Off-Center RWCM

Each grid point will have a cell mass

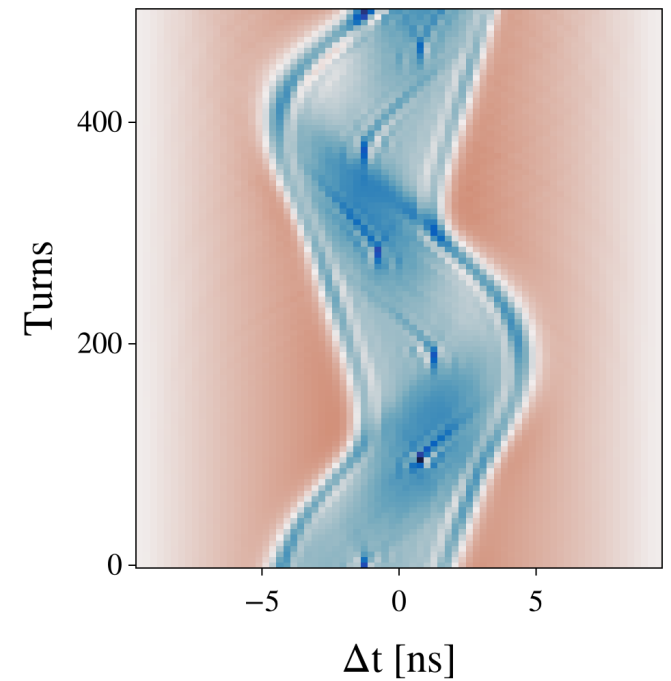
Ground-truth RWCM cascade



Reconstructed RWCM cascade



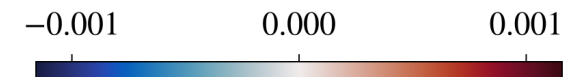
Profile residual



Normalized Line Density



Profile Residual

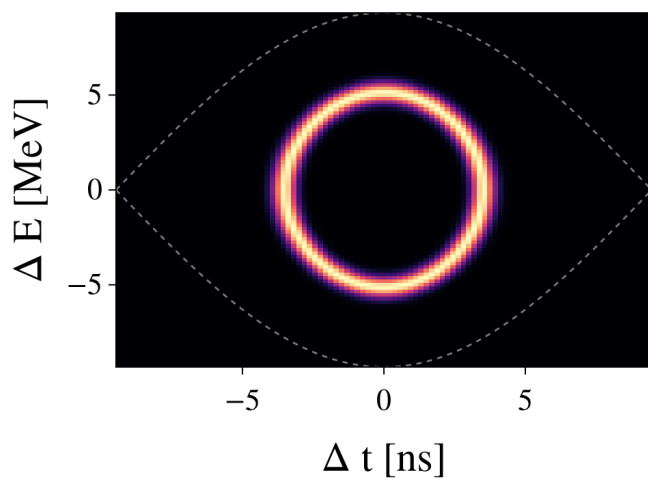




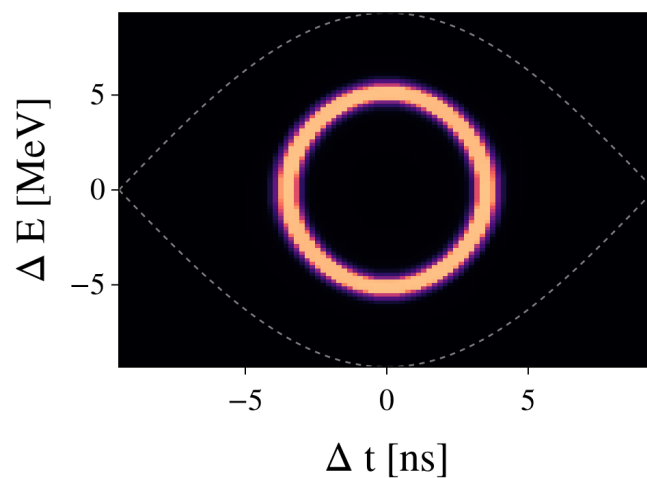
Airbag distribution

This is a self-consistent solution in longitudinal dynamics

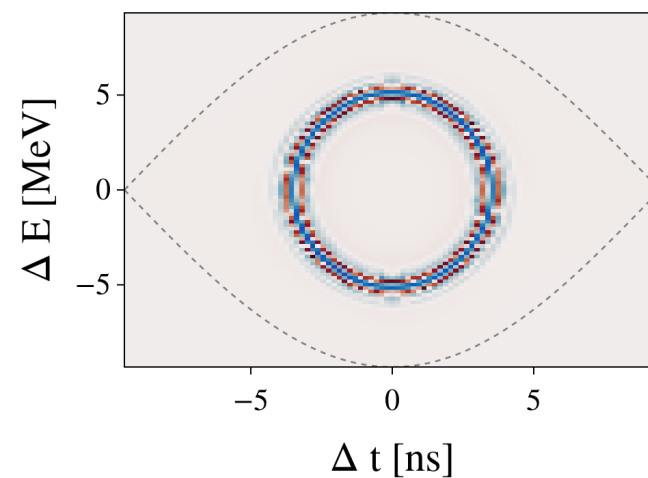
Ground-truth initial density



Grid-tomography reconstruction



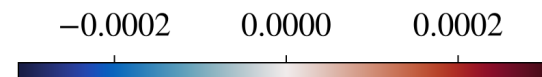
Residuals



Cell Mass



Cell Mass Difference

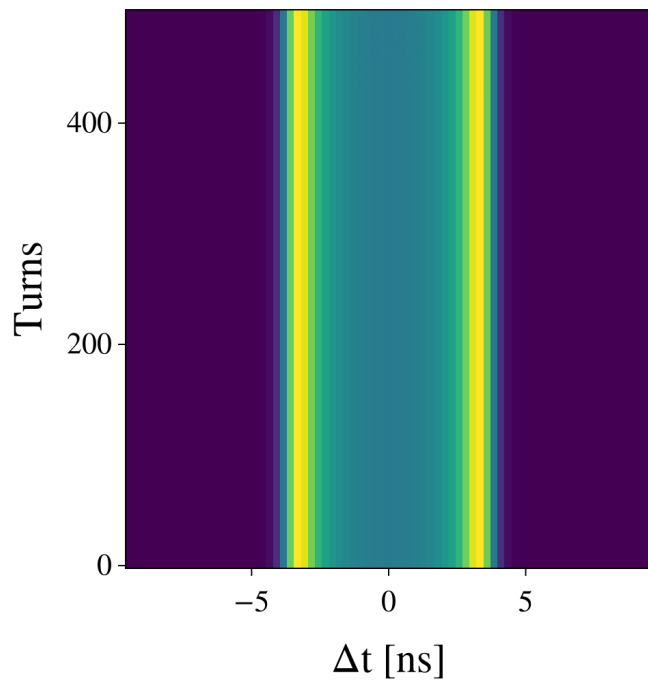




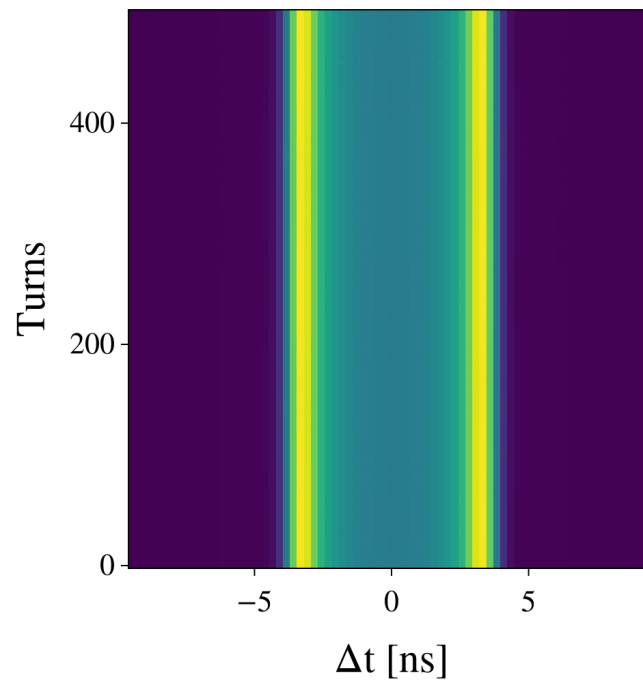
Airbag Distribution RWCM

Each grid point will have a cell mass

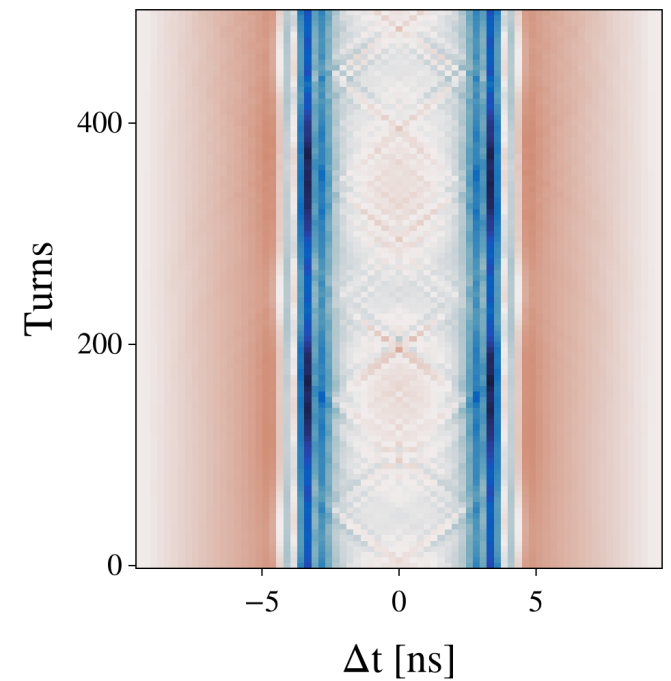
Ground-truth RWCM cascade



Reconstructed RWCM cascade



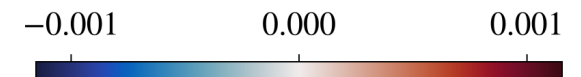
Profile residual



Normalized Line Density



Profile Residual

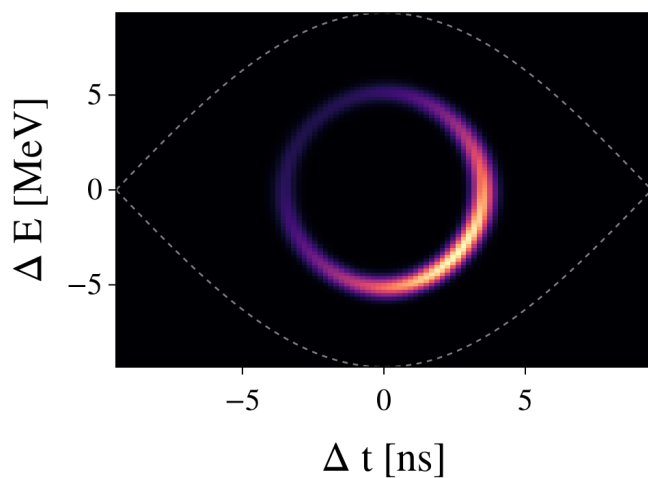




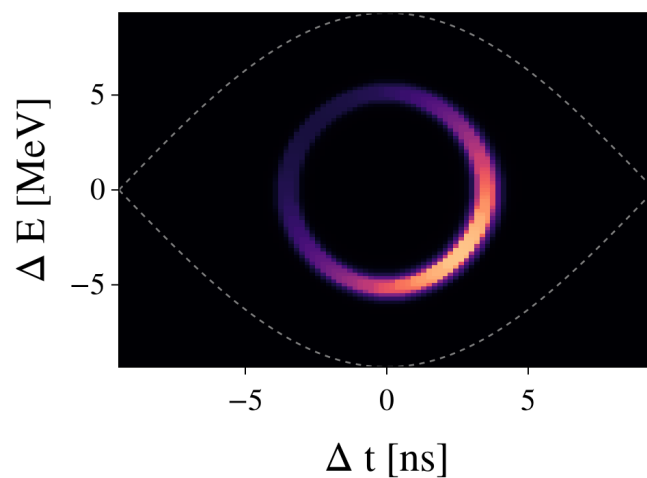
Solar Eclipse Distribution

Each grid point will have a cell mass

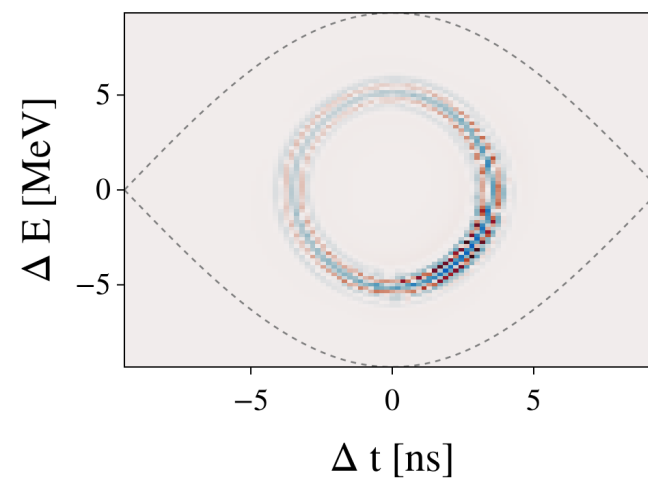
Ground-truth initial density



Grid-tomography reconstruction



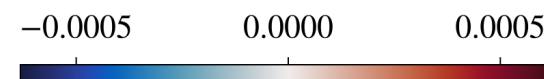
Residuals



Cell Mass



Cell Mass Difference

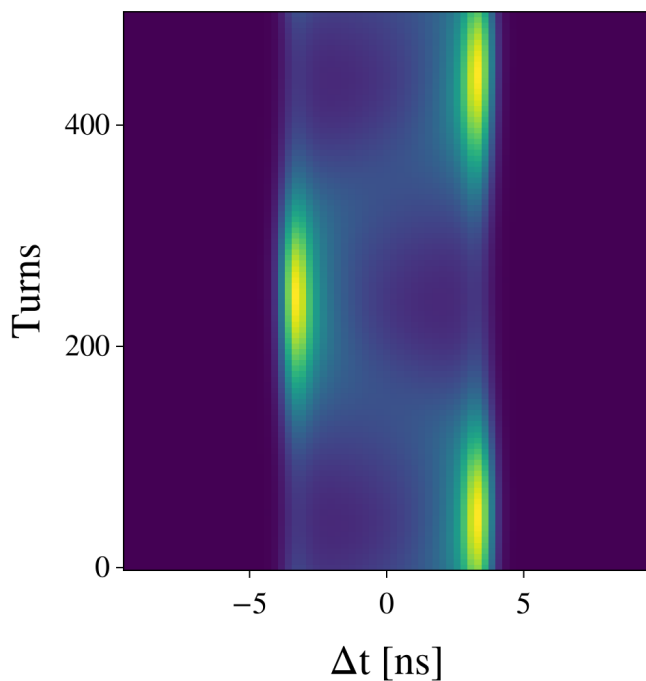




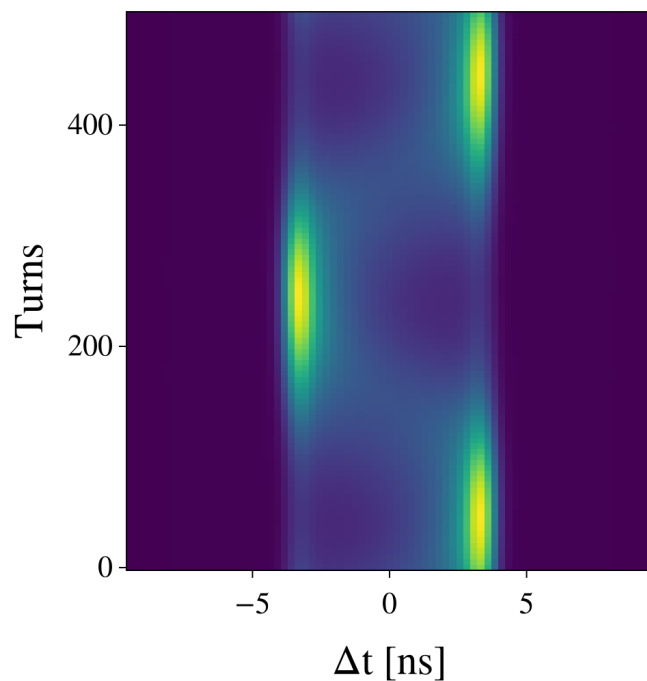
Solar Eclipse RWCM

Each grid point will have a cell mass

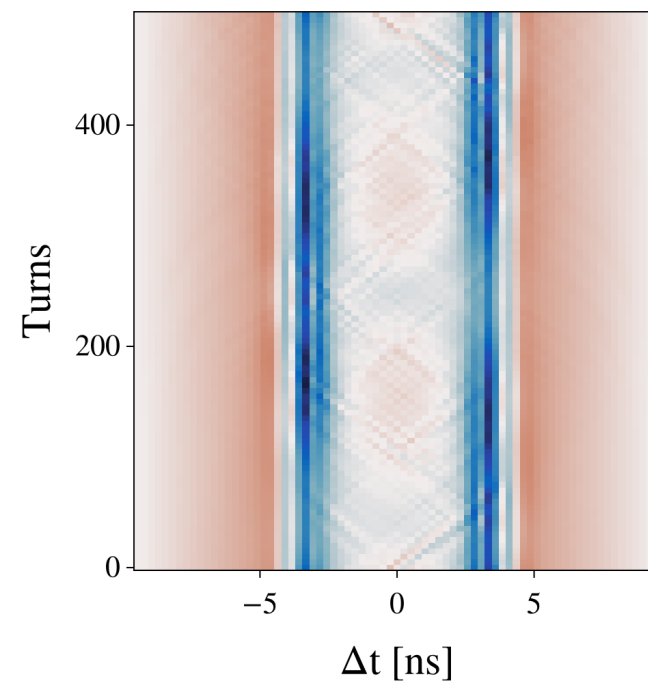
Ground-truth RWCM cascade



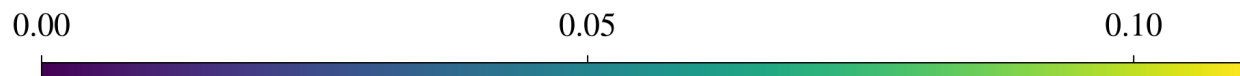
Reconstructed RWCM cascade



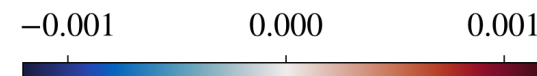
Profile residual



Normalized Line Density



Profile Residual

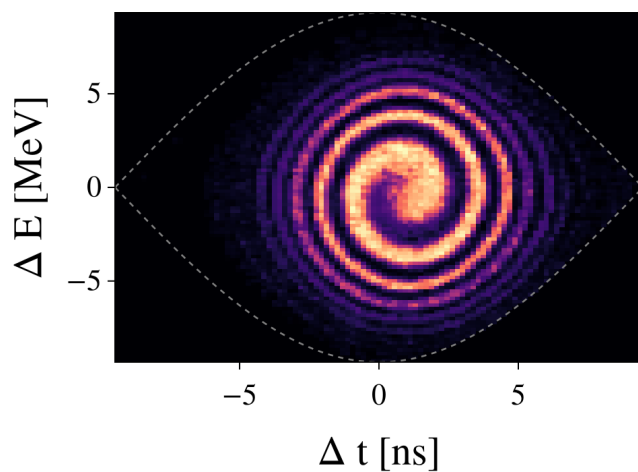




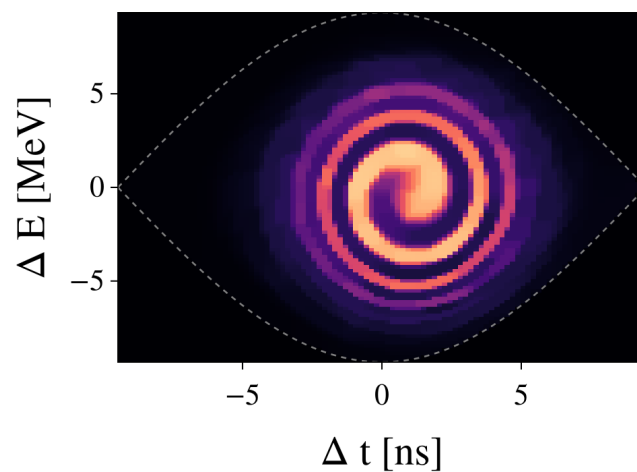
Spiral Distribution

Each grid point will have a cell mass

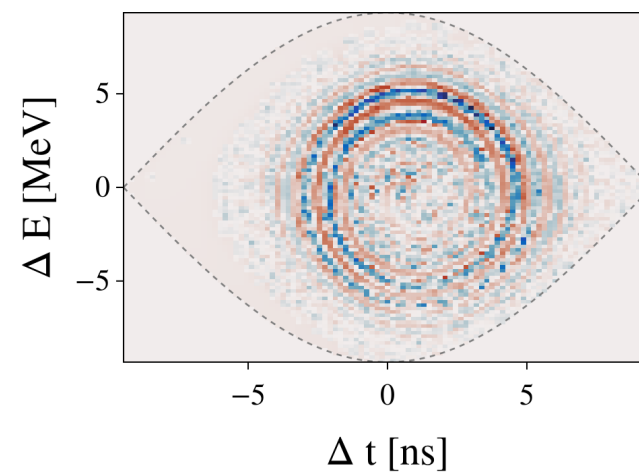
Ground-truth initial density



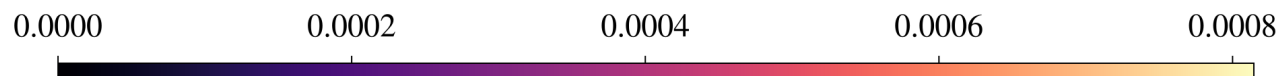
Grid-tomography reconstruction



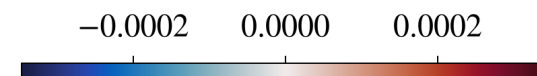
Residuals



Cell Mass



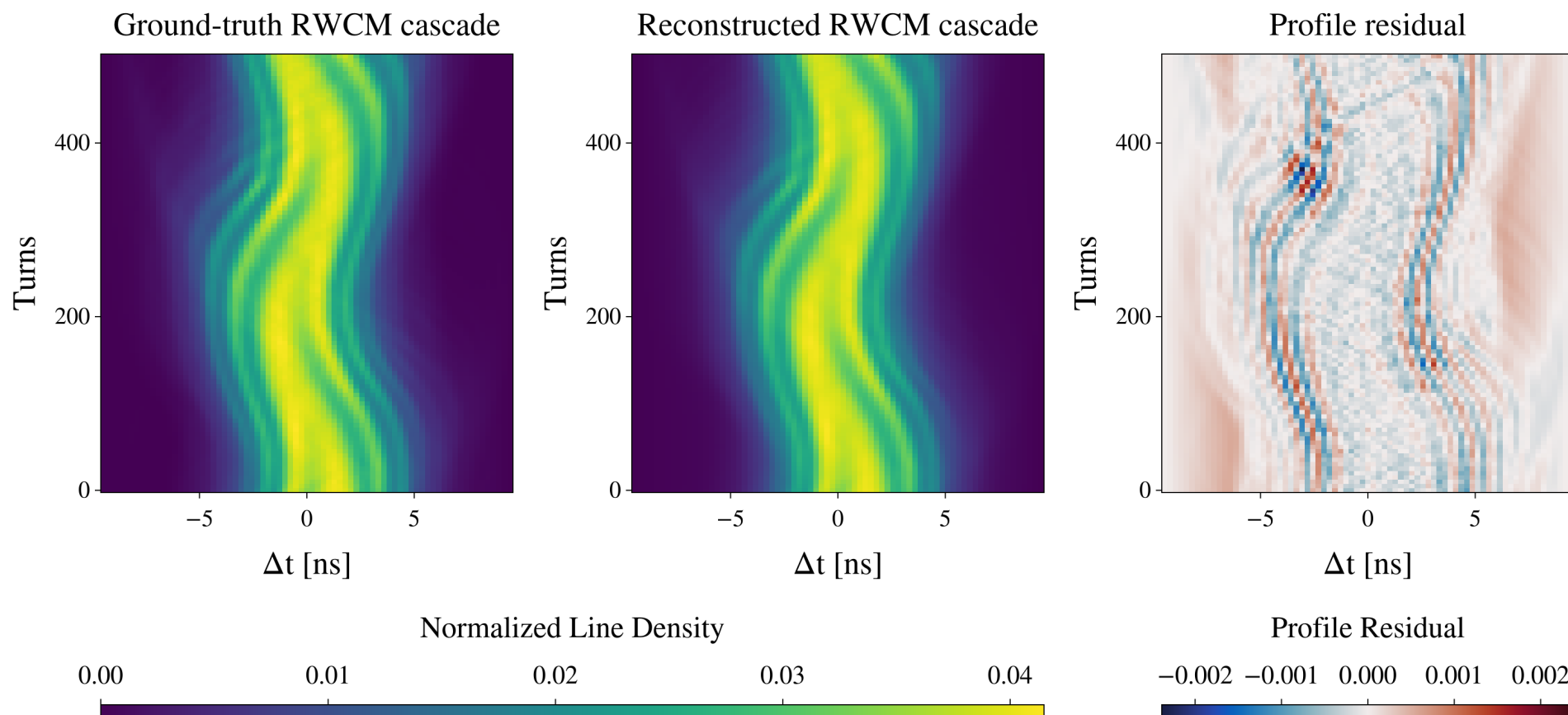
Cell Mass Difference





Spiral Distribution

Each grid point will have a cell mass

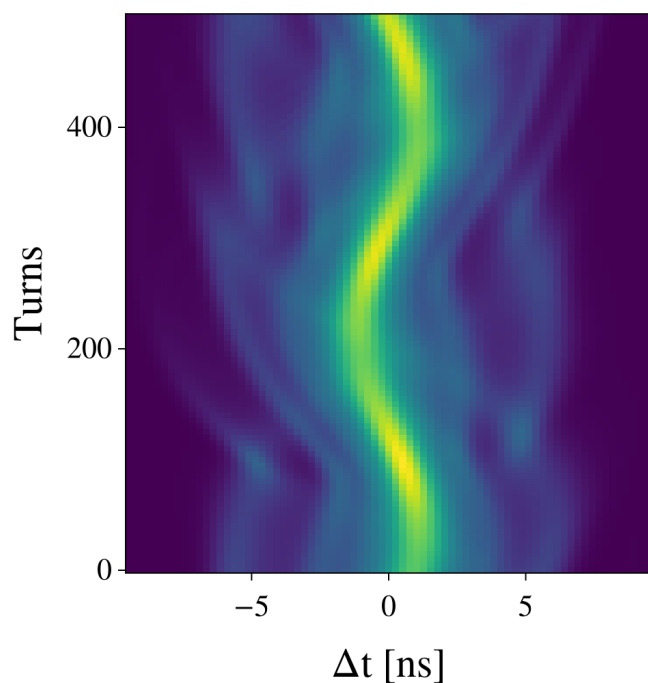




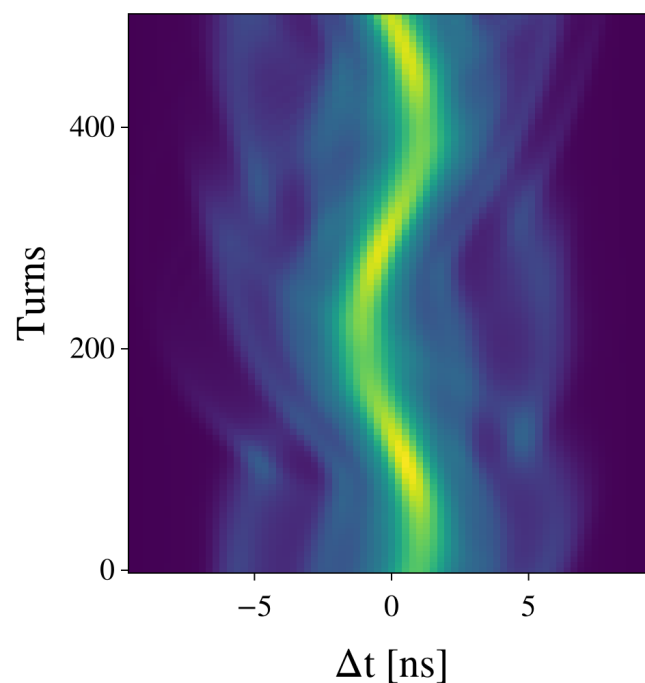
Spiral Distribution RWCM

Each grid point will have a cell mass

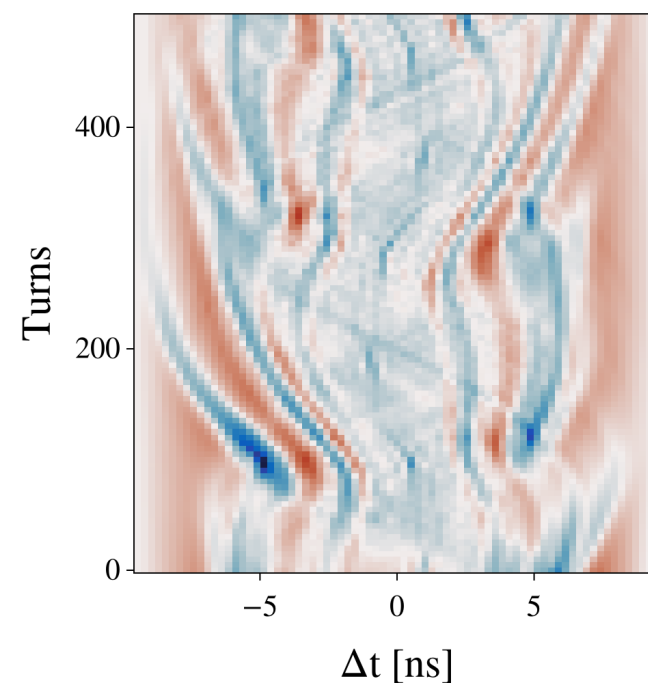
Ground-truth RWCM cascade



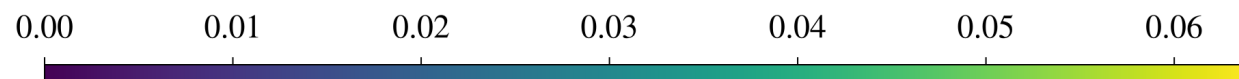
Reconstructed RWCM cascade



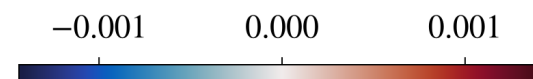
Profile residual



Normalized Line Density



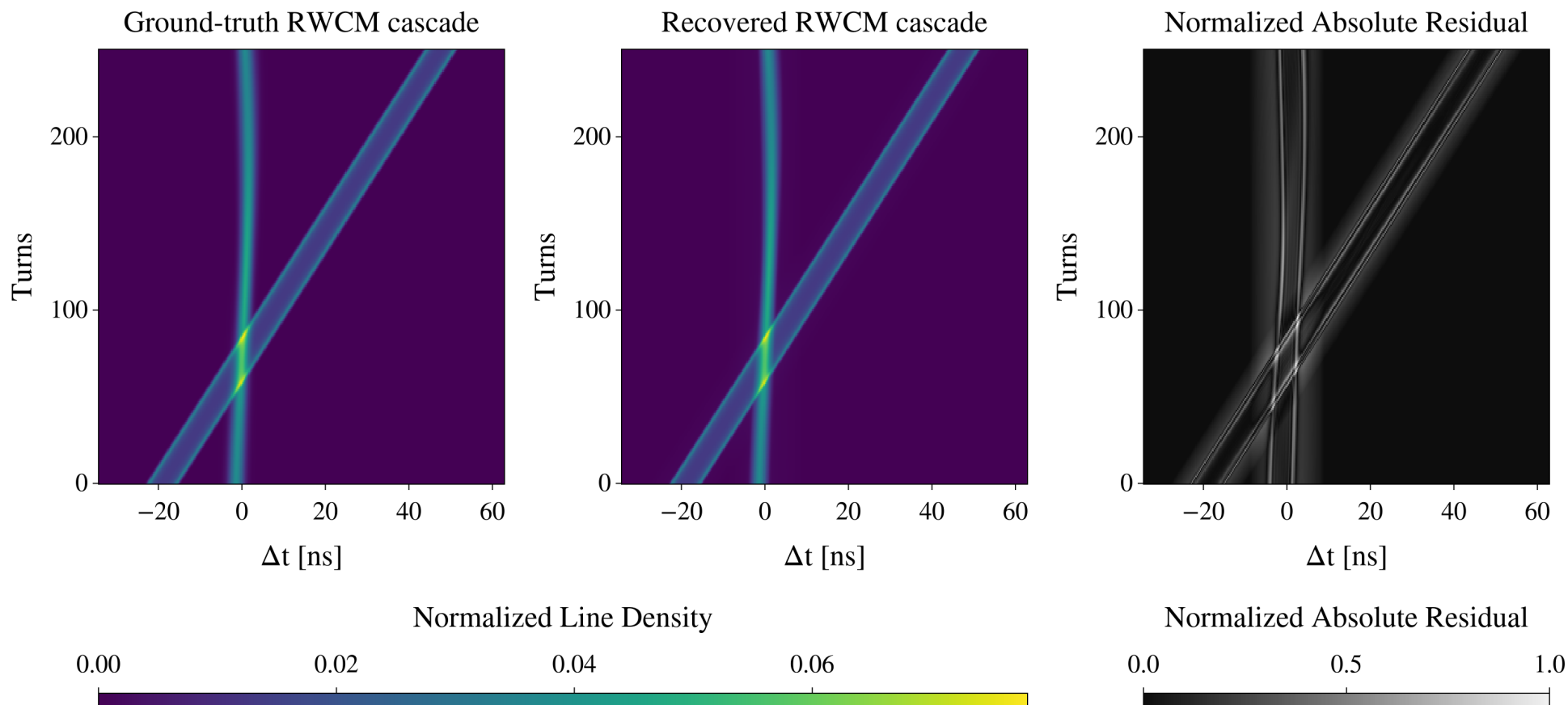
Profile Residual





Spiral Distribution RWCM

Each grid point will have a cell mass





Conclusions and Future Work

Proof of concept for grid tomography using analytical solutions has been made

**Include slip-stacking
and perturbations**

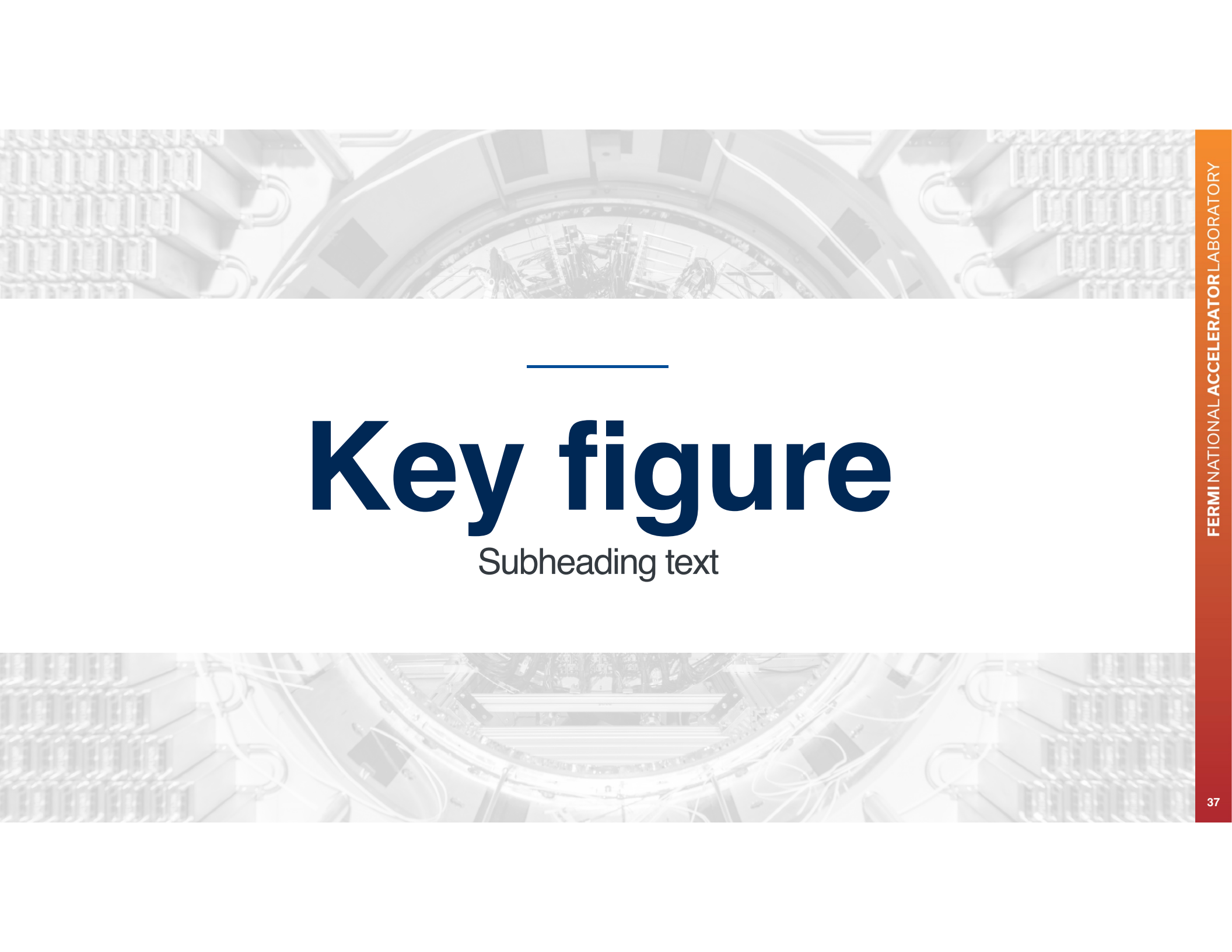
**Differentiable version
of BLOND (?)**

**Include machine
parameters in algorithm**

Refine loss functions

**Try on experimental
data**

**Include noise from
experiments**

The background of the slide is a grayscale photograph of the interior of a large particle accelerator tunnel. The tunnel is circular and filled with complex machinery, including various pipes, cables, and structural supports. The perspective is looking down the length of the tunnel, creating a sense of depth. The lighting is somewhat dim, with highlights on the metallic surfaces.

Key figure

Subheading text



Fermi**FORWARD**



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